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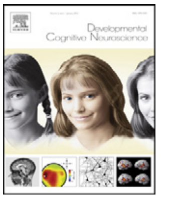

Neural circuitry associated with two different approaches to novel word learning

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ABSTRACT

Skilled reading depends upon successfully integrating orthographic, phonological, and semantic information; however, the process of becoming a skilled reader with efficient neural circuitry is not fully understood. Short-term learning paradigms can provide insight into learning mechanisms by revealing differential responses to training approaches. To date, neuroimaging studies have primarily focused on effects of teaching novel words either in isolation or in context, without directly comparing the two. The current study compared the behavioral and neurobiological effects of learning novel pseudowords (i.e., pronouncing and attaching meaning) trained either in isolation or in sentential context. Behavioral results showed generally comparable pseudoword learning for both conditions, but sentential context-trained pseudowords were spoken and comprehended slightly more quickly. Neurobiologically, fMRI activity for reading trained pseudowords was similar to real words; however, an interaction between training approach and reading proficiency was observed. Specifically, highly skilled readers showed similar levels of activity regardless of training approach. However, less skilled readers differentiated between training conditions, showing comparable activity to highly skilled readers only for isolation-trained pseudowords. Overall, behavioral and neurobiological findings suggest that training approach may affect rate of learning and neural circuitry, and that less skilled readers may need explicit training to develop optimal neural pathways.

1. Introduction

Reading is a foundational life skill, and as such has been extensively studied. To be considered a skilled reader, an individual must be able to recognize and comprehend written words efficiently (Shankweiler, 1999), employing

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a highly specialized system that integrates orthographic, phonological, and semantic information (e.g., Coltheart et al., 2001; Plaut et al., 1996). Indeed, a rich literature exists on normal and abnormal reading development and the implications for educational approaches (for review, see Rayner et al., 2001), and it is clear that a necessary component for developing appropriate educational practices for teaching students to read is understanding the mechanisms by which one learns to read. Although the need for brevity limits an in-depth discussion, a common theme across models of reading development is the relative importance of orthographic, phonological, and semantic information during learning and how the role of each changes across development, especially semantics (Coltheart et al., 2001; Ehri, 2005; Plaut et al., 1996). In particular, word recognition research based upon computational and phase/stage models suggests that early in reading development, the mapping of orthography onto phonology is central; however, over time and with increasing reading skill, the role of phonology diminishes whereas the role of semantics increases (Coltheart et al., 2001; Ehri, 2005; Plaut et al., 1996). Moreover, the importance of semantics is further illustrated by its differential role in the acquisition of exception words (i.e., showing greatest influence on inconsistent vs. consistent words; McKay et al., 2005; Juel, 1980; Stanovich and West, 1983; Strain et al., 2002; Taylor et al., 2011).

In compliment to the rich literature on the behavioral aspects of reading, neurobiological research has advanced our knowledge of the neural systems involved in reading. In particular, findings have not only revealed the brain regions implicated in reading but how these regions work together and, at the gross level, appear to map onto behavioral models of reading. Studies utilizing fMRI of readers engaged in single word reading have identified a left-hemisphere network including temporo-parietal (angular and supramarginal gyri), inferior frontal, and occipito-temporal cortices; these brain regions show differential engagement across various types of tasks as well as different stages of development (e.g., Fiez and Petersen, 1998; Mechelli et al., 2003; Pugh et al., 1996; Shaywitz et al., 2004; Turkeltaub et al., 2003). More specifically, both inferior frontal and temporo-parietal cortex (supramarginal gyrus) have a stronger response to pseudowords and low frequency words than to high frequency words (Fiez and Petersen, 1998; Katz et al., 2005; Simos et al., 2002a; Xu et al., 2001), implicating these regions in phonological processing and mapping phonology to orthography (Sandak et al., 2004a). Areas within the temporo-parietal cortex have also been suggested to play a role in the mapping of semantic knowledge (angular gyrus; Pugh et al., 1996; Sandak et al., 2004a). Thus, within the broad region of temporo-parietal cortex, a division of labor may exist with supramarginal gyrus more involved in phonological processing and angular gyrus more involved in semantic processing (Price et al., 1997; Sandak et al., 2004a). A final component of the left-hemisphere network important for reading is within occipito-temporal cortex, which is associated with fast and efficient word reading with the functional specificity shown to develop later (Sandak et al., 2004a; Shaywitz et al., 2002, 2004). Taken together, extant findings suggest that inferior frontal and temporo-parietal cortex appear to dominate early word reading acquisition (Turkeltaub et al., 2003) and are important for orthographic and phonological mapping, but with increasing proficiency a transition to greater reliance on occipito-temporal cortex occurs for fast and efficient word recognition (Sandak et al., 2004b; Shaywitz et al., 2002).

Nevertheless, how individuals *develop* an underlying efficient neural system for reading is less well understood, and as such remains to be an active area of research. Longitudinal neuroimaging research has reported evidence of "normalization" of neural systems after long -term remediation in poor readers (Shaywitz et al., 2004; Eden et al., 2004; Simos et al., 2002b). Such knowledge of the neurobiological correlates of learning, particularly differential effects for various types of learners, has direct implications for evidence-based development of educational approaches to reading instruction. While such long-term studies provide valuable information about the plasticity of neural systems as related to intervention, use of short-term learning paradigms allow investigators to efficiently design shorter-term, well-controlled experiments to test different training approaches. In this way behavioral and neuroimaging studies using short-term learning paradigms can help elucidate how teaching approach may interact with the learner's cognitive profile and the type of information to be learned, prior to conducting a large scale interventions (e.g., McKay et al., 2005; Pugh et al., 2008; Sandak et al., 2004b; Taylor et al., 2011). Indeed, findings from studies support that such approaches are viable. For example, using behavioral methods only, some short-term learning studies have examined the influence of semantics on word consistency and have found that semantic pre-exposure leads to faster and more accurate reading of novel words, particularly for inconsistent novel words, revealing important aspects of the relationship between semantic information and orthography (McKay et al., 2005; Taylor et al., 2011). Differences between learning words in isolation versus context have also been shown and reveal that the former is a better teaching approach for less skilled readers, while for skilled readers approach matters less (e.g., Landi et al., 2006).

Studies using behavioral *and* neuroimaging methods combined have provided insight not only into the behavioral correlates of short-term learning, but also the underlying neural mechanisms associated with the learning approach (e.g., Frishkoff et al., 2010; Sandak et al., 2004b; Pugh et al., 2008). For example, Sandak et al. (2004b) used a short-term learning paradigm to investigate changes in the reading network associated with novel word learning. During training, participants were shown pronounceable pseudowords and were asked to attend to one of three stimulus characteristics: orthographic, phonological, or semantic. Following training, participants were asked to read aloud three sets of printed stimuli during fMRI: trained pseudowords; untrained pseudowords; and high- and low-imageability low-frequency words. Behaviorally, superior performance was seen for the phonological and semantic training conditions; similar patterns were observed functionally. Specifically, there was no effect of orthographic training; however, within occipito-temporal cortex there was evidence of increased efficiency (relatively reduced activity) for phonological training, suggesting that occipito-temporal cortex is also sensitive to the phonological structure of words (Sandak et al., 2004b). Overall, these results provide evidence that activity within the reading network can be modulated by training method.

Other work has shown neurobiological effects of short-term training by embedding words in sentences (e.g., Borovsky et al., 2010; Frishkoff et al., 2010). For example, in a recent ERP study, Frishkoff and colleagues trained novel words under two sentential contexts, one providing strong semantic cues whereas the other was uninformative. After a delay (2 days) participants were exposed to a semantic priming task. There was evidence for differential N400 effects, such that familiar words and those trained within a strong semantic context elicited stronger N400 effects, whereas novel and words trained with an uninformative context elicited weaker N400 effects. These findings contribute to the rich literature that shows the beneficial role of semantics in word recognition, and is in line with the findings of a transition in older children and adults to greater reliance upon incidental learning via texts (Jenkins et al., 1984; Nagy et al., 1985).

Although neurobiological short-term paradigms focused on training words in either isolation or in context have been conducted with skilled adult readers, to our knowledge a direct comparison has yet to be made between these training conditions. Such a comparison is potentially important, as the behavioral literature provides a strong foundation for suggesting interactions between learning approaches and type of reader. Specifically, while adult skilled readers appear to benefit from learning new words via contextual training, beginning and less able readers benefit from learning words explicitly and in isolation (Lyon et al., 2003). In fact, research findings suggest that this distinction exists even within the more limited range of average versus above average readers: as mentioned above, Landi et al. (2006) showed that those with superior skills learn new words regardless of approach, whereas those that are simply in the average range (yet not necessarily impaired) benefit from direct and explicit instruction. Thus, use of short-term training approaches to learn more about the neuroscience of reading has already begun to generate information that can be useful in the development of targeted instructional programming. Within this context, the aims of this study using a short-term learning paradigm were threefold:

(1) To examine whether adult readers differ in word learning and recognition abilities at the behavioral level depending on whether the to-be-learned stimuli were trained in (a) a sentential context or (b) by more explicit but less contextualized methods;
(2) To examine whether training approach would result in modulation of neural activity within critical reading-related regions; and
(3) To examine whether level of reading proficiency combined with training approach would result in further modulation of brain regions associated with reading.

Behaviorally, we hypothesized that both training conditions would lead to efficient word learning, but that subtle differences in the patterns of performance across our behavioral measures *during* training might exist. Specifically, due to findings reporting the facilitory effect observed for words learned/presented within sentences, we expected to see enhanced word reading and comprehension for words trained via our sentential context, particularly for highly skilled readers. Neurobiologically, we expected that while both conditions would activate regions within the reading network, trained words would show patterns of activity more similar to real words versus novel pseudowords, as has been reported previously; however, we predicted that some differences would be present between training approaches, even if both resulted in similar behavioral performance after training. Additionally, we hypothesized that the differential patterns of activity within the reading network for trained pseudowords could be related to the degree of reading competence, with increased neural efficiency for explicit training being more critical for individuals who were not as highly skilled readers.

2. Materials and methods

2.1. Participants

Thirty-four adults, 21 to 36 years of age, completed the study. Participants were recruited via flyers posted at the Johns Hopkins Medical Institute and Homewood campuses. All participants met the following criteria: (1) native English speaker, (2) normal/corrected-to-normal hearing and vision, (3) no history of developmental disability, major psychiatric illness, and/or neurological disorder, and (4) no contraindication to the MRI environment. Written consent was obtained and procedures were carried out in accordance with the Johns Hopkins Medical Institutional Review Board.

All participants were in the normal range for basic reading (BR) skills, as assessed by the Letter Word Identification and Word Attack subtests from the Woodcock-Johnson Psychoeducational Battery–III (WJ-III; Woodcock et al., 2000). A BR composite score at the 35th percentile or higher was required for further participation in the study. Of the 34 participants who met initial eligibility criteria, 6 were excluded due to either extremely low reading scores, missing data, and/or recording errors throughout the session, yielding 28 adults (13 males; mean 25.4 ± 3.6 years) in the final sample. To examine the impact of reading proficiency on training condition and the observed neural response (Question 3), participants were divided into two groups based upon a median split of the BR composite score (cf. Landi et al., 2006). For maximal separation of the groups, we created a "buffer zone" (Shankweiler et al., 1999). This resulted in nine participants being classified as Average readers (range of 36th to 56th percentile) and 14 classified as Excellent readers (range of 70th to 97th percentile). Participants whose scores fell within 60th through 69th percentile were excluded ($n = 5$). A χ^2 analysis indicated that there were no significant differences in the distributions of males and females within each group (Average:

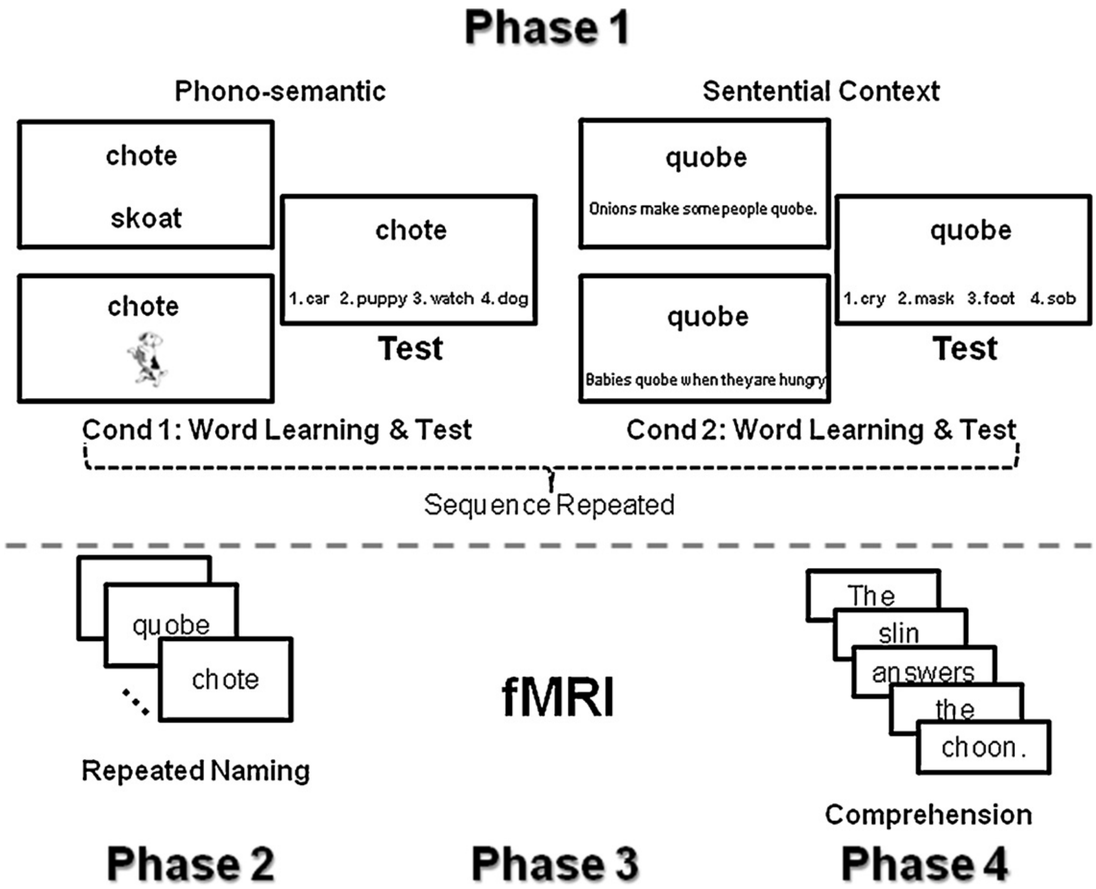

Fig. 1. Schematic illustration of the procedures for the behavioral and functional neuroimaging session. Examples of the phono-semantic and sentential context training conditions are provided. Participants completed the phases as illustrated.

$\chi^2(1) = 0.11$, $p = 0.74$; Excellent: $\chi^2(1) = 0.29$, $p = 0.59$) and between groups (Males: $\chi^2(1) = 0.40$, $p = 0.53$; Females: $\chi^2(1) = 0.69$, $p = 0.41$).

2.2. Materials, design, and procedures

Participants completed both behavioral and neuroimaging components. The behavioral component consisted of training and testing (phase 1), repeated naming (phase 2), and sentence comprehension (phase 4). Phases were completed in a fixed order to insure adequate learning/recognition prior to scanning (phase 1 and 2) and retention of associated word meaning after scanning (phase 4; Fig. 1). Across the different phases, when appropriate, stimuli were matched as close as possible on length, part of speech (1/2 nouns, 1/2 verbs), and imageability calculated via the MRC Psycholinguistics Database (Coltheart, 1981; see Table 1). All pseudoword stimuli were created by changing one letter of a real monosyllabic word not used elsewhere in the study.

2.2.1. Phase 1—word learning and test

To simulate novel word learning, participants were taught to associate meaning (synonyms) with 40 novel pronounceable pseudowords that adhered to the orthographic and phonological rules of English. Stimuli were split into two sets of 20 with one set used for phono-semantic (PS) training and the other for sentential context (SC) training (see Appendix A); training sets were counterbalanced across participants. Participants completed one learning set and corresponding test (e.g., PS) followed by the other learning set and corresponding test (e.g., SC); the entire sequence was then repeated. Order of conditions was counterbalanced across participants.

2.2.1.1. Phase 1 learning. The PS training required learners to pay attention to *both* phonological and semantic information (Sandak et al., 2004b,c). For this condition, participants were randomly presented with either a rhyming pseudoword or a picture; each presented three times (6

Table 1
Means and standard deviations for word length and imageability.

	Word Length	Imageability
Set 1 trained pseudowords	4.0 ± 0.8	561.4 ± 52.6
Set 2 trained pseudowords	4.1 ± 0.9	553.5 ± 131.7
High frequency words[a]	4.2 ± 0.7	463.2 ± 123.3
Low frequency words	4.2 ± 0.7	523.8 ± 82.8
Untrained pseudowords	4.2 ± 0.7	

[a] High frequency words are significantly different from the trained pseudoword sets.

exposures). The rhyming pseudoword required attention to the phonological aspects of the target, whereas the picture required attention on the associated meaning. For example, the stimulus "chote" was paired with the rhyming pseudoword "skoat" and with a picture of a dog (Fig. 1). At the start of training, participants were told that they would see new words and their job was to learn them. For each item, participants were instructed to read aloud the novel word and the rhyming word presented below it ("chote", "skoat") or to read the novel word and to label the accompanying picture ("chote", "dog"). If either of these were incorrect, participants were corrected and an incorrect response was recorded. Blocked presentation of the trials was used and the order of the blocks was random.

For the SC condition, participants were required to infer the intended meaning of pseudoword stimuli from the sentence contexts in which they were embedded. Each item was presented within two separate sentences; each paired three times (6 exposures). Trials were blocked and the order of the 20 blocks was randomly determined for each participant. Sentences were piloted to insure that the intended meaning could be inferred. At the start of training, participants were told that they were going to see words that they were supposed to learn. Participants were instructed to read aloud the novel word and the sentence that was printed below it, and then to state what the word meant ("quobe"; "onions make some people quobe"; "cry"). If any of the responses were incorrect, participants were corrected and an incorrect response was recorded.

2.2.1.2. Phase 1 test. To evaluate how well participants learned the association between pseudowords and their intended meanings (synonyms), a forced choice procedure was used after completion of each learning set. On each trial, one of the 20 pseudowords was shown at the top of the screen with four different word meanings listed below it (Fig. 1). Each set of choices included the following: correct; semantically related; previously trained meaning from same set; and randomly selected word not used elsewhere in the experiment. Participants were asked to read aloud the pseudoword, read aloud the choices, and to state the correct associated meaning. If the incorrect meaning was selected, the trial was marked as incorrect by the experimenter. For each condition (PS and SC), the test consisted of 40 trials (two repetitions of each pseudoword). Participants needed to demonstrate greater than 90% accuracy for both conditions on Test 2 before moving on to Phase 2, and no individuals failed to do so.

2.2.2. Phase 2—repeated naming

During repeated naming, the trained pseudowords were presented one at a time, and the participant read them aloud as quickly and accurately as possible. There were 20 randomizations of the list of 40 trained pseudowords, for 800 responses total. Accuracy and response time was recorded.

2.2.3. Phase 4—sentence comprehension

After scanning (Phase 3), participants were tested on their retention of the trained pseudoword meanings and the ability to comprehend them in a novel context. Participants read aloud sentences that were composed of both real words and pseudowords. Some sentences contained two trained pseudowords (from same condition) whereas others contained unfamiliar pseudowords (not used in other phases). Participants were asked to decide whether the sentence was plausible (e.g., meaningful) or not (e.g., not meaningful). Sentences were designed such that participants could only make judgments if retention of the intending meanings was maintained (were at chance when UPs used). For example, for the sentence, "The slin answers the choon", if the associated meaning of slin was "hair" and choon was "phone", then the sentence would be judged "not meaningful". The experimenter recorded the participant's response.

2.2.4. Phase 3—functional neuroimaging session

Immediately following Phases 1 and 2, participants underwent the scanning session, with the intended goal to examine recognition of the trained pseudowords and assess the degree to which the reading network was differentially engaged.

2.2.4.1. Design. In the lexical decision task, stimuli were presented on the screen, one at a time, and participants were asked to decide whether the letter string formed a real word. Participants were exposed to five stimulus types: trained pseudowords, untrained pseudowords, and real words (high and low frequency). Word frequency (established via Educator's Word Frequency Guide[1]; Zeno et al., 1995) was manipulated due to known differential responses seen within the reading network, enabling us to more directly assess the nature of the observed trained pseudoword activity. High frequency words corresponded to a standard frequency index (SFI) equal to or greater than 60, whereas low frequency words corresponded to a SFI equal to or less than 49.9. Participants were instructed to press the button in their right hand if the word was real, and press with their left hand if the word was not. Participants were told to treat the trained pseudowords as if they were real, so only the untrained pseudowords required a left-handed response.

The task consisted of 160 different stimuli; each run consisted of 80 stimuli randomly presented: 20 trained pseudowords (1/2 PS, 1/2 SC), 20 untrained pseudowords, and 40 real words (1/2 high frequency, 1/2 low frequency). Stimuli were presented on the screen for 1500 ms, with a jittered inter-stimulus interval averaging 2500 ms (range: 500–8000 ms). Stimuli were viewed through a mirror fixed on the MRI head coil and were projected by an LCD projector onto a rear-projection screen. The paradigm was computer-controlled with E-Prime (Psychology Software Tools, Pittsburgh, PA, USA), which presented the task and recorded the stimulus presentation time and participant response.

[1] Although not typically used for adults, this measure was chosen due to ongoing work with this paradigm examining possible differences between children and adults as well as between children who are typically developing readers and those identified as struggling with reading.

2.2.4.2. FMRI data acquisition and image analysis. Scanning was carried out in a 1.5 T ACS-NT Powertrack 6000 MRI scanner (Philips Medical Systems, Inc.) using an 8-element coil converted to a 6-channel head coil (for system compatibility). Functional MRI data were acquired using single shot echo planar imaging using sensitivity encoding with the following parameters: echo time = 40 ms, repetition time = 2 s, 128 × 128 matrix, and 240 mm field of view. Twenty-eight 4.0 mm thick axial slices with 1 mm inter-slice gap were collected providing whole brain coverage.

Image analysis was performed using Statistical Parametric Mapping 8 (SPM8; http://www.fil.ion.ucl.ac.uk/spm/) on Matlab R2009b (Mathworks, Inc., Natick, MA, USA). For each participant, rec images were converted to Analyze format, realigned, spatially normalized to Montreal Neurological Institute (MNI)-labeled space, re-sampled into $(3\,\text{mm})^3$ voxels, and smoothed using a $(8\,\text{mm})^3$ Gaussian kernel. Task associated activation was assessed using an event-related design and SPMs were created corresponding with the time-course for each of the five word types relative to a resting baseline, which consisted of the average response to the fixations between word presentations. Voxel-wise contrast maps for each subject were carried to a second-level analysis.

As a first pass, SPMs for each condition were entered into a one-sample t-test to examine significant activity associated with each condition. All SPMs were corrected using the family-wise error (FWE) rate and extent cluster threshold of 20 voxels. The local maxima of significant clusters were assigned neuroanatomic labels using the WFU_PickAtlas database (Maldjian et al., 2003). A repeated measures analysis of variance (RM-ANOVA) with condition as a within-subjects factor was used to examine patterns of activation associated with trained pseudowords as compared to real and untrained pseudowords. In order to address Question 2 (whether differences between training conditions existed), we specifically contrasted PS and SC.

Finally, region of interest (ROI) analyses were employed to examine whether more fine-grained differences within key reading-related regions were modulated by training approach and/or reading proficiency. Differences in neural activity across conditions were assessed in four regions taken from previous literature (Fiez and Petersen, 1998; Simos et al., 2002a; Xu et al., 2001): inferior frontal gyrus (IFG; BA 44/45), angular gyrus (AG), supramarginal gyrus (SMG), and occipito-temporal cortex (OTC) defined via Pickatlas. Both SMG and OTC were defined by 6 mm spheres centered on MNI coordinates $x = 42$, $y = -44$, $z = 40$ and $x = 42$, $y = -50$, $z = -18$, respectively, whereas IFG (BA 44/45) and AG were created via predefined masks within Pickatlas. Percent signal change was extracted using Marsbar (Brett et al., 2002) from each cluster and were entered into a mixed-ANOVA with condition as a within-subjects factor and group as a between-subjects factor.

3. Results

For all analyses, when appropriate, within-subject factors were corrected for nonsphericity using Greenhouse-Geisser corrected probabilities and post-hoc analyses were Sidak corrected for multiple comparisons.

3.1. Behavioral session

For each phase, accuracy (percentage correct) and/or response time (RT) were recorded (see Table 2). In addition to testing differences between training conditions during the learning process for the overall group (Aim 1), we also tested whether there was any additional modulation of these effects based upon reading proficiency. There was no significant effect of reading group throughout the behavioral session (except for one difference in sentence comprehension, noted in Table 3) including in scanner behavioral performance on the fMRI lexical decision task (see Table 4). Moreover, there was little change to the data with the inclusion of reading group as a between-subjects factor; therefore, only results from the overall group are reported below.

3.1.1. Phase 1—word learning and test

3.1.1.1. Phase 1 learning. Amount of word learning (% correct) was assessed via RM-ANOVA with condition and exposure (1st, 2nd) as within-subjects factors. There was a significant main effect of exposure, $F(1, 27) = 75.79$, $p < 0.001$ ($\eta_p^2 = 0.74$), indicating a reliable improvement in performance from the first to the second exposure. However, there was no significant main effect of condition, $F(1, 27) = 2.78$, $p = 0.11$ ($\eta_p^2 = 0.09$), or significant exposure × condition interaction, $F(1, 27) = 0.80$, $p = 0.38$ ($\eta_p^2 = 0.03$), indicating no difference in overall amount of learning between PS and SC.

3.1.1.2. Phase 1 test. To assess the strength of the learned word associations, we examined performance (% correct) during testing after each learning exposure via RM-ANOVA with condition and test time as within-subjects factors. There was a significant main effect of test time, $F(1, 27) = 93.09$, $p < 0.001$ ($\eta_p^2 = 0.78$), indicating a reliable improvement in performance from the first to the second test. However, there was no significant main effect of condition, $F(1, 27) = 0.159$, $p = 0.69$ ($\eta_p^2 = 0.01$) or significant test time × condition interaction, $F(1, 27) = 0.967$, $p = 0.33$ ($\eta_p^2 = 0.04$), indicating no difference in the amount of learning between the PS and SC.

3.1.2. Phase 2—repeated naming

Accuracy was extremely high, with all participants at or above 97%; therefore, no additional analyses were conducted. Mean RTs less than 200 ms and greater than 1500 ms were discarded as outliers. To examine the change in repeated naming across exposures, RT data for each condition were binned into equal quarters (5 time points each), and a RM-ANOVA was conducted with condition and time bin as within-subjects factors. A significant main effect of condition, $F(1, 27) = 5.36$, $p = 0.03$ ($\eta_p^2 = 0.17$), indicated that SC had reliably faster overall RT than PS. There was also a significant main effect of time bin, $F(3, 81) = 5.51$, $p = 0.01$ ($\eta_p^2 = 0.17$). Post hoc analysis revealed that the first time bin was significantly different from the second

Table 2
Means and standard deviations for the behavioral training session for the phono-semantic and sentential context conditions.

	Phono-semantic	Sentential context	$t(27)$	P
Phase 1 learning (% correct)				
1st exposure	97.2 ± 2.0	96.7 ± 3.1	0.98	0.34
2nd exposure	99.6 ± 0.8	98.7 ± 2.9	2.16	0.04
Phase 1 test (% correct)				
1st exposure	81.5 ± 10.5	83.3 ± 11.8	0.74	0.47
2nd exposure	96.3 ± 5.4	95.5 ± 5.8	0.74	0.47
Phase 2 rapid naming (ms)				
Overall	559.8 ± 112.9	553.4 ± 112.3	2.28	0.03
Time bin 1	586.1 ± 130.2	568.3 ± 126.3	3.89	0.001
Time bin 2	553.0 ± 117.7	547.8 ± 114.0	1.26	0.22
Time bin 3	550.6 ± 115.3	544.3 ± 110.1	1.03	0.31
Time bin 4	546.8 ± 108.4	548.6 ± 110.8	0.32	0.75
Phase 4 sentence comprehension				
Accuracy (% correct)	81.3 ± 16.3	85.5 ± 13.1	1.87	0.07
Response time (ms)	696.8 ± 148.6	678.1 ± 107.1	1.09	0.29

Table 3
Means and standard deviations for the behavioral training session for the phono-semantic and sentential context conditions broken down by reader group.

	Phono-semantic		Sentential context	
	Average	Excellent	Average	Excellent
Phase 1 Learning (% correct)				
1st exposure	96.6 ± 2.9	97.3 ± 1.5	96.8 ± 2.3	96.8 ± 3.4
2nd exposure	99.5 ± 1.1	99.7 ± 0.6	98.1 ± 3.5	98.9 ± 3.1
Phase 1 test (% correct)				
1st exposure	81.4 ± 10.6	83.9 ± 11.3	81.1 ± 13.1	84.8 ± 9.9
2nd exposure	95.8 ± 4.8	95.7 ± 6.6	95.0 ± 7.7	95.5 ± 4.9
Phase 2 rapid naming (ms)				
Overall	580.1 ± 85.2	536.6 ± 105.5	578.8 ± 92.4	529.8 ± 101.4
Time bin 1	608.6 ± 120.9	558.5 ± 136.9	593.1 ± 129.3	545.7 ± 117.5
Time bin 2	579.0 ± 92.9	553.0 ± 117.7	575.8 ± 93.1	522.5 ± 99.9
Time bin 3	571.9 ± 71.1	522.0 ± 94.3	577.6 ± 82.0	514.2 ± 91.6
Time bin 4	556.5 ± 77.6	528.3 ± 111.5	565.0 ± 78.3	531.5 ± 101.5
Phase 4 sentence comprehension				
Accuracy (% correct)	74.3 ± 15.5	84.8 ± 18.1	79.2 ± 12.9	87.5 ± 13.9
Response Time (ms)**	783.2 ± 189.2	621.6 ± 87.1	733.3 ± 99.9	620.0 ± 82.4

** There was a significant difference between groups on overall RT for Phase 4 sentence comprehension test, $F(1, 21) = 9.69$, $p = 0.01$, with excellent readers generally faster than average readers.

and third ($ps = 0.04$), with no other differences observed. Importantly, there was a significant condition × time bin interaction, $F(3, 81) = 2.79$, $p = 0.05$ ($\eta_p^2 = 0.09$). As shown in Fig. 2, this can be attributed to the greater change in RT over time bins for PS as evidenced by a significant decrease in response latency over time, whereas SC resulted in more consistent response time from start to finish. Overall, results suggest that the main effect of condition and its interaction with time are largely due to the growth observed in PS.

3.1.3. Phase 4—sentence comprehension

Overall, participants performed fairly well with performance greater than 80%. There was no difference between conditions in overall RT, $t(27) = 1.09$, $p = 0.29$; however, there was a suggestion of a slight advantage for SC over PS

Table 4
Means and standard deviations for accuracy (% correct) and response time (ms) broken down by reader group and overall performance.

	Accuracy (%)			Response time (ms)		
	Average	Excellent	Overall	Average	Excellent	Overall
High frequency words	98.4 ± 3.5	99.5 ± 2.0	99.1 ± 2.4	672.6 ± 54.6	690.1 ± 50.1	699.2 ± 62.5
Low frequency words	96.2 ± 4.3	98.2 ± 3.3	97.7 ± 3.5	748.4 ± 51.1	752.2 ± 68.2	763.8 ± 65.7
Sentential context	90.0 ± 8.5	86.1 ± 13.9	86.9 ± 12.7	773.9 ± 56.3	791.7 ± 64.4	792.5 ± 65.2
Phono-semantic	83.5 ± 17.7	85.4 ± 13.7	83.7 ± 14.9	816.3 ± 58.1	819.3 ± 101.5	827.0 ± 81.3
Untrained pseudowords	84.2 ± 8.6	87.5 ± 7.3	85.5 ± 7.5	903.4 ± 75.1	887.7 ± 75.5	909.9 ± 79.0

Notes: It should be noted that the overall accuracy and response time are slightly divergent from the average of the Average and Excellent reader groups; this is due to the fact that there are 5 participants that were excluded when conducting the reader group analysis.

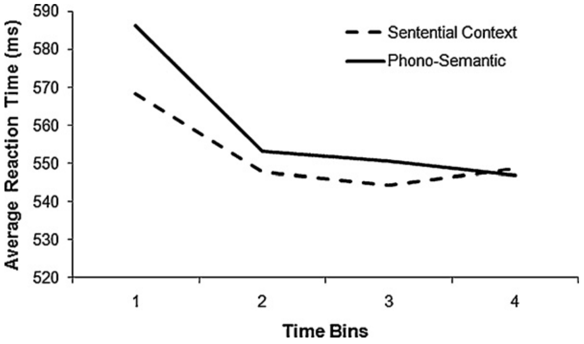

Fig. 2. Mean reaction time for the rapid naming task broken down by training condition and time bins. The time bins consist of 20 exposures to each word, broken down into equal length bins (5 exposures = 1 bin).

in accuracy, $t(27) = 1.87$, $p = 0.07$, possibly indicating that having learned words within a sentence may facilitate recognition and/or comprehension.

3.2. MRI task performance

Task accuracy (% correct) and mean RT were measured for each condition[2] and were entered into separate RM-ANOVAs with condition as a within-subjects factor. A summary of the overall results, as well as results broken down based upon reading group, can be found in Table 4.

Overall, real words had higher accuracies and were significantly different from trained pseudowords (both SC and PS) and untrained pseudowords, $F(4,100) = 17.76$, $p < 0.001$ ($\eta_p^2 = 0.09$), which was confirmed by post-hoc analysis, (all ps < 0.01); no other differences between conditions was observed (all ps > 0.9). For RT, there was a significant main effect of condition, $F(4,100) = 50.92$, $p < 0.001$ ($\eta_p^2 = 0.09$), with high frequency words yielding the fastest RTs, followed by low frequency words, both trained pseudowords, and finally the untrained pseudowords. Post-hoc analysis revealed differences amongst all conditions (all ps < 0.01) except between SC and PS ($p = 0.29$) and between SC and low frequency words ($p = 0.59$).

3.3. fMRI data analysis

3.3.1. Whole-brain analysis

Results of the one-sample t-tests for each condition (Fig. 3) revealed a common network of activity across stimulus types associated with visual processing including bilateral primary visual cortex extending into ventral extrastriate (BA 18/19) and posterior temporal cortex (BA 20/37). Activity was also seen in parietal cortex including AG and SMG (BA 39/40). A network of activity including bilateral cerebellum, primary motor cortex, and pre-/post-central gyrus was seen: left hemisphere activity was seen for trained pseudowords and real words consistent with a correct right-hand button press, whereas right hemisphere activity was seen for untrained pseudowords. Additional activity within left frontal cortex for real words and trained pseudowords was seen in IFG. Trained pseudowords also showed activity in dorsolateral prefrontal cortex and supplementary motor cortex extending into anterior cingulate. Together, these results suggest that in general there were similarities in the networks of regions activated across word types, but that some differences were seen.

3.3.2. Word type contrasts

Direct comparisons (RM-ANOVA) amongst conditions indicated that both real words and trained pseudowords showed significantly greater activity than untrained pseudowords in left inferior parietal cortex (AG), precuneus/posterior cingulate, and motor cortex. An additional area of increased activity was found in orbitofrontal and anterior cingulate cortex for real words as compared to untrained pseudowords, which was driven primarily by deactivation for untrained pseudowords. In terms of differences between trained pseudowords and real words, trained pseudowords showed greater activity in left IFG, inferior parietal cortex (AG), and a small region within precuneus. Overall, our results show that both trained pseudowords and real words showed comparable differences from untrained pseudowords, suggesting that trained pseudowords were functioning similarly to real words; however, trained pseudowords also had areas of significant difference from real words, suggesting that some distinctions existed, possibly modulated by training condition and/or reading ability.

3.3.3. Trained pseudoword contrasts (PS and SC)

When comparing PS and SC directly, no differences were observed at the more stringent FWE correction; however, when a less stringent cluster-based threshold was applied ($p < 0.001$ uncorrected, extent of $k = 78$, still equivalent to $p < 0.05$), greater activity in left AG, extending into middle temporal cortex, was seen for PS as compared to SC.

3.3.3.1. ROI analyses. Fig. 4 summarizes the observed patterns of activity across conditions by reader type for each ROI. For IFG, there was a non-significant main effect of group, $F(1,21) = 0.46$, $p = 0.51$ ($\eta_p^2 = 0.02$), but a significant main effect of condition, $F(4,84) = 4.46$, $p = 0.005$ ($\eta_p^2 = 0.18$). Post hoc analysis showed that real words and trained pseudowords were all significantly different from untrained pseudowords (all $p \leq 0.05$); additionally, low frequency words were significantly different from high frequency words ($p = 0.03$), and PS was marginally significantly different from high frequency words ($p = 0.06$). More importantly, however, there was a significant condition × group interaction, $F(4,84) = 2.64$, $p = 0.05$ ($\eta_p^2 = 0.11$); an interaction contrast of PS vs. SC by Average Readers vs. Excellent Readers was significant, $F(4,84) = 7.58$, $p < 0.001$ and accounted for 83% of the interaction. Simple effects analysis showed that Average Readers had significantly greater activity for PS than SC ($F(1,84) = 11.55$, $p < 0.001$), whereas there was no significant difference between trained pseudowords for Excellent Readers ($F(1,84) = 0.04$, $p = 0.85$). These results indicate that the PS condition for the Average Reader group was most similar to the levels of activity seen for trained pseudowords for Excellent Readers. Moreover, it should be

[2] Two participants (1 male) had missing data due to hardware problems.

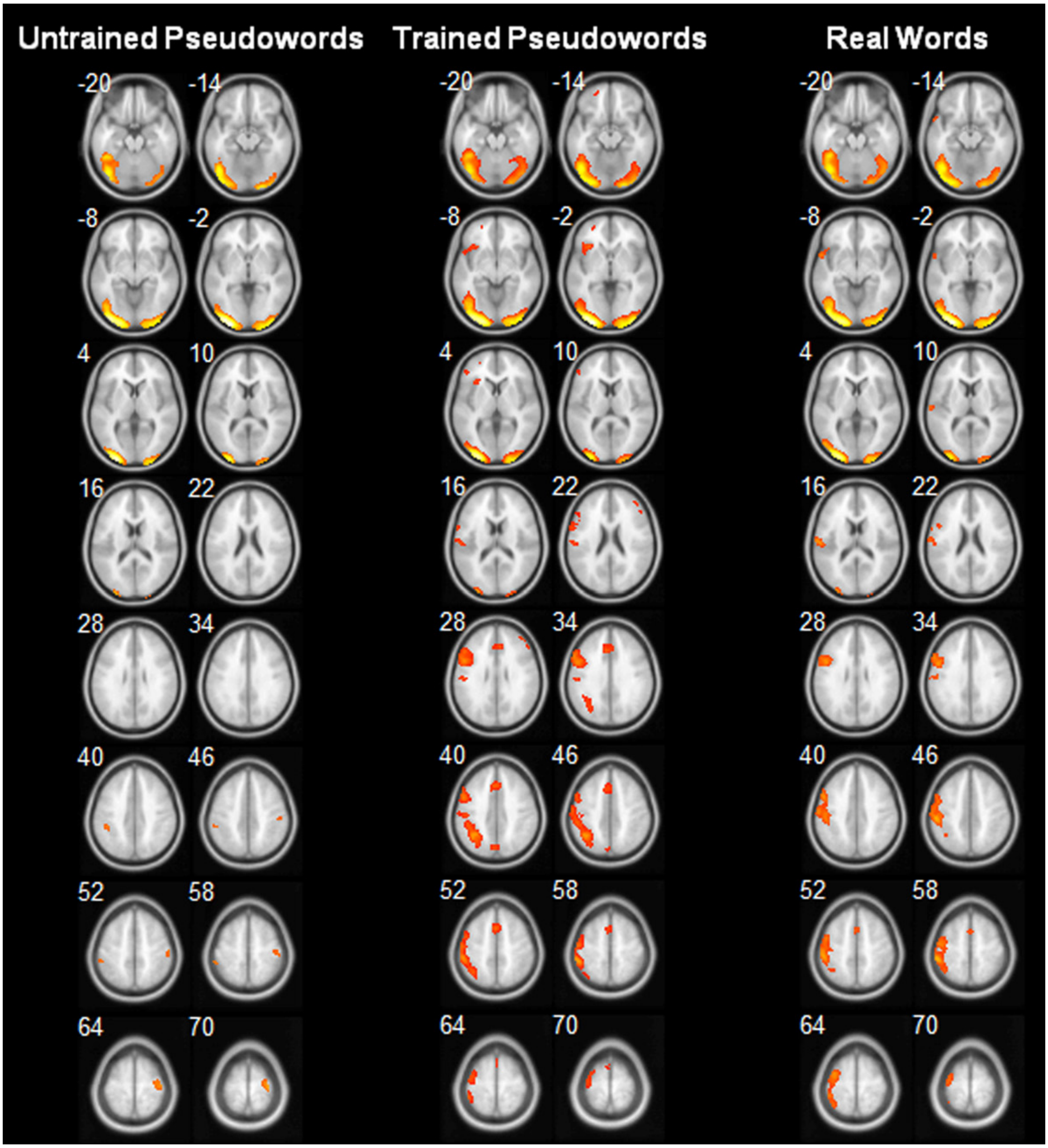

Fig. 3. Statistical parametric maps of one-sample *t*-tests for activity associated with untrained pseudowords, trained pseudowords, and real words greater than baseline (fixation). Trained pseudowords and real words are collapsed across conditions. Regions in yellow/red represent significant activation. Images are in neurological convention (Left = Left).

noted that this region overlaps with the significant difference seen between trained pseudowords and real words in the whole-brain contrasts. Taken together, evidence suggests that reading proficiency modulates neural activity for trained pseudowords leading to the significant difference observed in the whole-brain data between trained pseudowords and real words.

For OTC, there was a significant main effect of condition, $F(4,85) = 3.97$, $p = 0.01$ ($\eta_p^2 = 0.16$). Post hoc analysis showed that all real and trained pseudowords were significantly different from untrained pseudowords ($ps < 0.05$), with no other differences found ($ps > 0.1$). There was no significant main effect of group, $F(1,21) = 2.32$, $p = 0.14$ ($\eta_p^2 = 0.10$), or interaction, $F(4,85) = 2.14$, $p = 0.11$ ($\eta_p^2 = 0.09$). Thus, reading proficiency showed no differential effect on trained pseudowords.

For SMG, there was a significant main effect of condition, $F(4,85) = 3.30$, $p = 0.02$ ($\eta_p^2 = 0.14$). Post-hoc analysis showed that PS and untrained pseudowords were significantly different from high frequency words ($ps < 0.05$); no

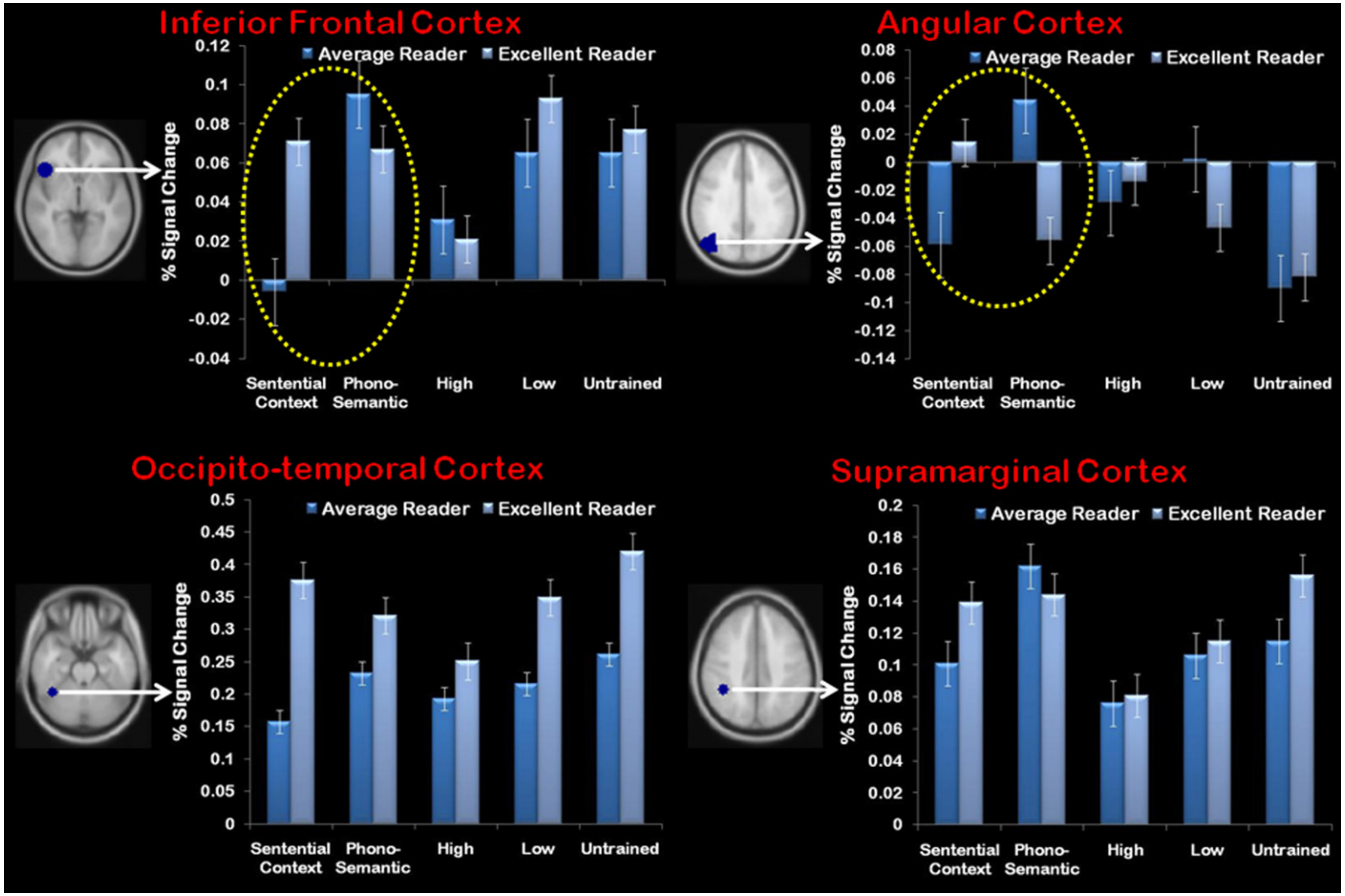

Fig. 4. Region of interest analysis summary results broken out by reading group for each region separately. The blue area for each brain image roughly corresponds to the center of each region. Percent signal change was averaged across all voxels within each region for each condition. The yellow circle highlights the significant interaction seen in IFG and AG. Images are in neurological convention (Left = Left).

other differences existed ($ps > 0.3$). There was no significant main effect of group, $F(1,21) = 0.11$, $p = 0.74$ ($\eta_p^2 = 0.01$), or interaction, $F(4,85) = 0.64$, $p = 0.60$ ($\eta_p^2 = 0.03$), suggesting that reading proficiency had little effect on trained pseudowords within this region.

Lastly, for AG, there was no significant main effect of group, $F(1,21) = 0.07$, $p = 0.79$ ($\eta_p^2 = 0.004$) and a marginally significant main effect of condition, $F(4,84) = 2.53$, $p = 0.07$ ($\eta_p^2 = 0.11$) which was accounted for by all real and trained pseudowords being different from untrained pseudowords. Importantly, there was a significant condition × group interaction, $F(4,84) = 2.83$, $p = 0.05$ ($\eta_p^2 = 0.12$). A contrast of PS vs. SC by Average Readers vs. Excellent Readers was significant, $F(4,84) = 6.63$, $p < 0.001$, accounting for 83% of the interaction. Simple effects analysis showed a similar pattern to that seen in the IFG: that Average Readers had significantly greater activity in PS as compared to SC ($F(1,84) = 4.00$, $p = 0.05$), whereas no significant differences between trained pseudowords emerged for Excellent Readers ($F(1,84) = 2.67$, $p = 0.11$). As noted above in the IFG results, this region showed significant overlap with the difference observed in the whole-brain data between trained pseudowords and real words. Thus, these results suggest that reading proficiency had a differential effect on trained pseudowords within this region and accounts for the difference observed in the whole-brain data between real words and trained pseudowords.

4. Discussion

This study was designed to examine whether skilled readers would show differential behavioral and neurobiological patterns for word learning that were associated with training approach. Participants learned to pronounce and attach meaning to pseudowords during behavioral training that was intended to simulate novel word learning under two conditions: phono-semantic, combining phonological and semantic aspects of training described by Sandak et al. (2004b,c), and sentential context. The lexical decision task administered during functional neuroimaging permitted comparisons of the patterns of activation for the newly learned pseudowords with those for real words and novel pseudowords.

In general, our results supported our predictions that we would see some subtle differences between conditions in the overall learning of the words. We found that there was a reliable improvement from the first test to the second test for both conditions, suggesting general word learning. However, when participants needed only to say the words aloud as quickly as possible or to read and comprehend sentences that used the words, we saw a slight advantage for words that were trained within a sentential context. This was evidenced by faster reaction times for the initial trials of rapid naming (although the phono-semantic condition did reach similar levels after five repetitions), and a suggestion of slightly better performance during the sentence comprehension task. The finding of an advantage for the sentential context condition is consistent with previous literature with adult readers that shows a faciliatory effect of word learning in context (Frishkoff et al., 2010; Perfetti et al., 1979; Stanovich, 1980; Stanovich and West, 1983). Thus, although participants were performing at essentially equal levels at the end of the behavioral training, our findings suggest a slight differentiation between learning phases that was dependent upon training approach. Although these findings on the surface suggest that sentential context training may be a superior approach compared to phono-semantic training, our neurobiological findings, particularly as they interacted with reading skill, suggest a more complex picture.

Neurobiologically, we had hypothesized that regardless of training condition, we would see activation in traditional reading-related regions (left occipito-temoporal, inferior frontal, and temporo-parietal cortex) and that the activity would be more similar to real words than novel pseudowords. The use of a lexical decision task provided us with some evidence as to how participants were treating these words. For example, if participants were treating the trained pseudowords as untrained pseudowords, then we would expect to see similar levels of activity in regions that process semantic information (within areas of temporo-parietal cortex); however, if participants are treating them as real words, then we should see similar levels of activity between real and trained pseudowords and this should be different from untrained pseudowords.

When examining the overall patterns of activity associated with our task, we saw that across all conditions there was significant activity in bilateral extrastriate extending into posterior temporal cortices. This activity has significant overlap with occipito-temporal cortex, one of the key reading-related regions (Fiez and Petersen, 1998; Graves et al., 2010; Herbster et al., 1997; Katz et al., 2005; Price et al., 2003; Pugh et al., 1996; Simos et al., 2002a; Tarkiainen et al., 1999; Xu et al., 2001). In general, we did not see evidence for a significant modulatory effect of our training conditions or reading proficiency in this region. Others have shown distinct patterns associated with different training approaches; for example, Sandak et al. (2004b), who compared phonological-only, semantic-only, and orthographic-only training showed that their phonological-only training condition resulted in increased efficiency compared to orthographic-only and semantic-only training. Of course, while we cannot exactly compare these findings to our study because both of our conditions had a semantic emphasis, Sandak et al. (2004c) also reported preliminary results comparing phonological- and semantic-only training to a combined phonological and semantic training (as utilized in the present study). Results of this preliminary study (Sandak et al., 2004c) showed that the combined condition yielded the most robust behavioral advantage, yet only showed an intermediate response within OT as compared to the other training conditions. These findings suggest that a combined phonological-semantic emphasis may be advantageous in terms of overall word learning, but does not yield as robust neural efficiency in OT. Both of our training conditions emphasized semantics, and although they showed increased neural efficiency compared to pseudowords, we did not find a robust advantage of phono-semantic training over sentential context training in OT efficiency. Taken together, this may indicate that adding a meaning component during learning of words may dampen OT response; whether this ultimately proves to be disadvantageous or advantageous remains to be seen. It may be fruitful for future studies to focus on relative emphases of phonological versus semantic weighting during training; an optimal weighting may exist that produces both efficient semantic learning and word recognition (reflected in OT signal).

We also found significant left temporo-parietal cortex activity across conditions, including angular and supramarginal cortices (BA 39/40); activity within this region has been associated with mapping orthography to phonology and semantic knowledge (McDermott et al., 2003; Pugh et al., 1996; Sandak et al., 2004b). For supramarginal cortex, we found a significant difference between the phono-semantic condition and untrained pseudowords from high frequency words, suggesting increased phonological processing demands for the former two, which would be consistent with previous literature (Price et al., 1997; Xu et al., 2001). No other differences amongst conditions and no modulation associated with reading proficiency were seen in supramarginal cortex.

In contrast to supramarginal cortex, for angular cortex, we found significant differences amongst our conditions. Specifically, there was an overlapping region where trained pseudowords and real words differed from untrained pseudowords, indicating that the trained pseudowords had similar levels of activity to real words. This is consistent with the thought that this region contributes to semantic processing (Pugh et al., 1996; Sandak et al., 2004b), as the untrained pseudowords were the only stimuli that did not have a mapped semantic association. There was also a small region within temporo-parietal cortex where trained pseudowords were significantly different from real words; analyses revealed a significant interaction between training approach and reading proficiency. Specifically, no differences in activity between training approaches were observed for the excellent readers, whereas significant differences were observed between training conditions for average readers; this suggests that the average readers may have formed a greater semantic association for the phono-semantic condition than the sentential context condition, whereas the excellent readers showed no differential response.

Another interpretation of our results in angular cortex may be that instead of calling upon the learned semantic association, participants may be relying upon an episodic memory trace. Previous ERP research on word learning has investigated the P600 component in posterior regions (e.g., parietal cortex) – a marker for recognition memory – has shown that there is a stronger effect for trained words with no effect for untrained and familiar words (Perfetti et al., 2005). Moreover, Perfetti and colleagues found a differential effect of word learning dependent upon reading proficiency such that participants characterized as strong comprehenders learned more words than their less skilled counterparts did and showed a stronger episodic memory effect. Although our study did not find a difference in overall word learning and reading proficiency as was found in the Perfetti study (2005), we cannot rule out that some of the observed results could be related to participants drawing upon an episodic memory trace.

Lastly, an additional aspect of our findings within the angular cortex is that we saw activity associated with deactivation (as compared to baseline; Fig. 4); this is consistent with previous research that have also shown deactivation related to semantic processing for active tasks similar to our lexical decision task (Binder et al., 1999, 2005; McKiernan et al., 2003). Deactivation has been hypothesized to reflect either reallocation of neural response or inhibition of neural response (Harel et al., 2002), although there is not yet a clear consensus regarding interpretation of deactivation.

Within frontal cortex, we saw patterns of activity for inferior frontal cortex supporting the role of this region in phonological processing, as has been suggested by previous literature (Fiez and Petersen, 1998; Graves et al., 2010; Price et al., 1997; Sandak et al., 2004a; Simos et al., 2002a). Although there was a significant difference between trained pseudowords and real words seen within the whole-brain analysis, our findings suggest that this was due in large part to the interaction between training condition and reading proficiency. Specifically, as seen in angular cortex, we found no difference between conditions for excellent readers, whereas average readers showed differential activation for the phono-semantic and sentential context conditions, with the phono-semantic condition showing the most similar level of activation as trained pseudowords for the excellent readers.

Another finding was additional activity within the frontal cortex for our trained pseudowords conditions that was seen in the whole brain results. Previous studies have shown that activity within left dorsolateral prefrontal cortex and supplementary motor cortex/anterior cingulate could be associated with increased demands needed for decision-making as well as motor preparation and execution (Carreiras et al., 2007; Elliott et al., 1999; Tremblay and Gracco, 2010; Walton et al., 2004). Therefore, this activation suggests that when participants saw the trained pseudowords, they were actively deciding if this was a word that was previously learned with a semantic association, and mapping this onto the correct motor response. Moreover, responses within these regions could be an artifact of the demands of the lexical decision task; therefore, future studies using tasks with reduced cognitive demands such as silent naming (as used by Sandak et al., 2004b,c) may help to provide a more pure measure of response to trained pseudowords.

Overall, this study using a short-term learning paradigm contributes to a better understanding of the neuroscience of reading as well as potentially providing useful groundwork for educational intervention studies. Generally, our results provide additional evidence for distinctive roles within the reading-related brain network, specifically, angular cortex for semantic access and inferior frontal cortex (and supramarginal cortex) for phonological processing. Moreover, the finding of an interaction between reading proficiency and training conditions sets the stage to extend this neurobiological research to different populations of readers and is directly relevant to educational settings and educational research. More specifically, our results for the phono-semantic condition in average readers showed specific neural correlates within inferior frontal cortex and angular cortex that was comparable to both trained pseudoword conditions for the excellent readers. The neurobiological findings may suggest that a more explicit training approach for less -skilled readers is beneficial, whereas for excellent readers, training approach has fewer differential effects. Such an interpretation is in line with behavioral findings that indicate that instructional approach to reading has less impact on better readers, whereas a more explicit approach clearly benefits less skilled readers (Foorman et al., 1998; Landi et al., 2006). It should be noted, however, that most reading intervention studies to date have not focused as much on the role of semantic training; our study suggests a potentially important interaction between reading skill and approach to training with regard to semantics. Further research with a wider range of reading skill will need to be conducted to fully unpack the implications of these findings.

Additionally, it will be important to further investigate whether training approach has differential effects on type of reading difficulty; for example, individuals with fluency difficulties may particularly benefit from sentential context training, whereas those with decoding difficulties may need more explicit and direct instruction. Specifically, future studies are needed to investigate changes within particular reading disabled populations, and whether these types of short-term learning paradigms can reveal more about the malleability of reading skills, especially for uncovering whether one training or educational approach might be most beneficial for students with different reading profiles.

It should be noted that other studies have examined the role of increasing delay in world learning and have shown increased facilitation in learning after a period of off-line consolidation (i.e., a week; Davis et al., 2008; Dumay and Gaskell, 2007). Therefore, we may not be capturing the entire learning process as distinctions could still emerge between training conditions as well as amongst word types with increasing delay; this will be an important area for further research. Moreover, while studies have examined the relative importance of semantics on learning of consistent vs. inconsistent orthography (McKay et al., 2005; Taylor et al., 2011), future work could incorporate the type of design utilized in this study to further explore explicit vs. more contextualized approaches to learning consistent vs. inconsistent words. Additionally, while no sex differences were observed in the current study overall and between reader groups, previous research has shown sex-based differences in language processing (for review see Hyde and Linn, 1988) as well as a suggestion of a disproportionate sex difference for reading disabilities (Flannery et al., 2000; Liederman et al., 2005 although see Shaywitz et al., 1990). Thus, additional research is needed to fully explore the relationship between sex differences, reading proficiency, and how this might map to differences in learning approaches. In particular, sex differences may contribute to a better understanding of the male-predominance of the epidemiology of dyslexia and lead to targeted instruction designed to capitalize on relevant neurobiological sex-differences.

In summary, our findings suggest that: (1) subtle differences can be detected, even in adult readers, during the learning process; (2) neurobiologically, trained pseudowords generally "function" similarly to real words; and (3) level of reading proficiency can influence the neural response within the reading network to the type of training employed. Thus, this work lays the foundation for more targeted investigations of what types of training may work for particular readers (in particular studies including a broader range of reading abilities to include impaired readers) and how this will translate into an efficient neural network. The eventual goal for this knowledge is to provide educators with the information necessary to design and utilize instructional approaches targeted to the neurobiological differences of different types of learners.

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Appendix A.

List A				List B			
Phono-semantic		Sentential context		Phono-semantic		Sentential context	
barp	girl	brip	key	vint	girl	swoy	key
quobe	cry	chote	dog	glant	cry	meest	dog
choon	phone	glant	climb	plute	phone	quobe	climb
meest	cat	nade	monkey	chote	cat	slin	monkey
fope	write	heke	read	jate	write	zale	read
gleeb	fly	kay	bear	troil	fly	nur	bear
hain	cut	jate	ski	maip	cut	fope	ski
jord	wash	grite	swim	lorp	wash	preet	swim
krile	sleep	dirn	kick	smope	sleep	teef	kick
lut	game	sneal	hand	fim	game	bufty	hand
nisk	bird	lorp	drive	gilk	bird	jord	drive
preet	throw	fim	car	grite	throw	lut	car
bufty	foot	plute	apple	sneal	foot	choon	apple
slin	hair	reen	duck	nade	hair	trun	duck
swoy	book	maip	run	brip	book	hain	run
teef	eat	smope	cheer	dirn	eat	krile	cheer
trun	mouse	troil	dance	reen	mouse	gleeb	dance
nur	cow	gilk	tree	kay	cow	nisk	tree
whum	sing	yab	build	yab	sing	wum	build
zale	drink	vint	boy	heke	drink	barp	boy

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Update

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Erratum

Erratum to: Supplement: Neuroscience & Education

In the 2012 *Developmental Cognitive Neuroscience* supplement 'Neuroscience & Education' (Volume 2, Supplement 1, ppS1-S204) the conflict of interest statements were missing from the following papers. The authors' conflict of interest statements are listed below.

Early adversity and neural correlates of executive function: Implications for academic adjustment

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Brain function during probabilistic learning in relation to IQ and level of education

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Neural circuitry associated with two different approaches to novel word learning

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Early math achievement and functional connectivity in the fronto-parietal network

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Weak task-related modulation and stimulus representations during arithmetic problem solving in children with developmental dyscalculia

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Neuronal effects following working memory training

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Practice effects in the developing brain: A pilot study

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Enhanced efficiency of the executive attention network after training in preschool children: Immediate changes and effects after two months

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